ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 1 · practice d.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 C ✓ 1.1.3 A ✓ 1.1.4 D ✓ 1.1.5 B ✓ 1.1.6 B ✓ 1.1.7 A ✓ 1.1.8 D ✓ 1.1.9 C ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ (5)

1.3 1.3.1 offline programming ✓ 1.3.2 cycle time ✓ 1.3.3 firmware ✓ 1.3.4 servo motor ✓ 1.3.5 AGV ✓ (5)

[20]

TOTAL SECTION A: [20]

SECTION B

QUESTION 2

2.1.1 $t = 20 \div 8$ ✓ $= 2{,}5$ h ✓ (2)

2.1.2 $E = 24 \times 20$ ✓ $= 480$ Wh ✓ (2)

2.2 Any TWO advantages ✓✓: high energy for their mass (light); long cycle life; fast charging; low self-discharge; no memory effect. Any TWO disadvantages ✓✓: expensive; need a battery management system; fire risk if damaged, overcharged or short-circuited; lose capacity in very hot or cold conditions. (4)

2.3 An AGV follows a fixed route (lines, tape, wires or markers) and stops when something blocks its path ✓✓. An AMR uses sensors (LiDAR, cameras) and a map to plan its own route and drive around obstacles ✓. (3)

2.4 Any TWO ✓✓: driving onto an automatic docking / charging station with contacts; wireless (inductive) charging pads; automatic battery-swap stations; opportunity charging at stops along the route. (2)

[13]

QUESTION 3

3.1 The suction cups are pressed onto the smooth glass ✓; a vacuum pump or venturi (ejector) removes air from inside the cups ✓; the higher atmospheric pressure outside pushes the cups onto the glass, holding it until air is let back in ✓. (3)

3.2.1 $A = \pi \times r^2 = \pi \times 0{,}05^2$ ✓ $= 0{,}00785\ m^2$; $F = p \times A = 60\,000 \times 0{,}00785$ ✓ $= 471{,}24$ N ✓ (3)

3.2.2 $W = m \times g = 60 \times 9{,}8 = 588\ N$; required force $= 2 \times 588 = 1\ 176\ N$ ✓; number $= 1\ 176 \div 471{,}24 = 2{,}5$ ✓; therefore 3 suction cups ✓ (always round up) (3)

3.3 Any TWO types with a suitable use ✓✓ each, e.g. two- or three-finger mechanical gripper – boxes, machine parts; magnetic gripper – steel sheets; soft (flexible) gripper – fruit and eggs; needle gripper – fabric and foam; adhesive gripper – light, thin materials. (4)

[13]

QUESTION 4

4.1 Serial communication sends the bits one after the other over one data line ✓; parallel communication sends several bits at the same time over several lines ✓. (2)

4.2 4.2.1 I2C: two wires (SDA data, SCL clock) ✓; e.g. connecting several sensors such as an IMU, display or real-time clock ✓. 4.2.2 SPI: four wires (MOSI, MISO, SCK, CS / SS) ✓; e.g. SD card, fast displays, ADC chips ✓. 4.2.3 UART: two wires (TX, RX) ✓; e.g. GPS module, Bluetooth module, link to a computer ✓. (6)

4.3 Data rate $= 2\ 000 \times 16$ ✓ $= 32\ 000$ bit/s ✓ $= 32$ kbit/s ✓ (3)

4.4 A protocol is a set of rules for how data is formatted, sent, received and checked ✓; if the devices use different protocols they cannot understand each other's data ✓. (2)

4.5 Any THREE ✓✓✓: damaged or loose cables and connectors; electrical noise from motors and VSDs (poor shielding / earthing); wrong IP addresses or settings; faulty switch or network card; wireless interference or poor signal; cables too long; wrong termination resistors on a fieldbus. (3)

[16]

QUESTION 5

5.1 A delta robot has three (or four) light arms joined to one moving platform above the work area ✓; the arms are driven by motors on a fixed base, which gives very fast, accurate movement ✓. Application ✓: fast pick-and-place of sweets, food or medicine packs; sorting small parts; packaging. (3)

5.2 $x = 0{,}5 \cos 30° + 0{,}3 \cos 90°$ ✓ $= 0{,}433 + 0$ ✓ $= 0{,}43$ m ✓; $y = 0{,}5 \sin 30° + 0{,}3 \sin 90°$ ✓ $= 0{,}25 + 0{,}3$ ✓ $= 0{,}55$ m ✓ (6)

5.3 $F = m \times a = 4 \times 3$ ✓ $= 12\ N$ ✓ (2)

5.4 The work envelope is the whole space that the robot's end effector can reach ✓. It is important because all the parts, machines and conveyors must be inside it ✓, and fences, guards and people must be kept outside it for safety ✓. (3)

5.5 Any THREE ✓✓✓: constant, high-quality welds; faster production; keeps workers away from fumes, heat, sparks and bright light; works long hours without tiring; less wasted material and rework. (3)

[17]

QUESTION 6

6.1 A digital input has only two states, on or off (1 or 0) ✓, e.g. limit switch, push button, PIR or inductive proximity sensor ✓. An analogue input can have any value in a range ✓, e.g. LDR, potentiometer, temperature sensor, pressure sensor ✓. (4)

6.2 $\mathrm{V} = \text{reading} \times \mathrm{V_{ref}} \div (2^{\mathrm{n}} - 1) = 614 \times 5 \div 1\,023$ ✓✓ $= 3{,}00\ \mathrm{V}$ ✓ (3)

6.3 PWM switches the supply fully on and off very quickly ✓, and the average voltage is changed by changing the on-time (duty cycle) ✓. At 25% the supply is on for a quarter of each period, so the motor gets about a quarter of the full voltage and runs slowly ✓. (3)

6.4 An optocoupler passes the signal with light (LED and phototransistor) ✓ so there is no electrical connection between the two circuits ✓; this protects the microcontroller from high voltages, voltage spikes and electrical noise from the motor circuit ✓. (3)

[13]

QUESTION 7

7.1 Any FOUR ✓✓✓✓: test at low speed first; keep clear of moving parts and the work envelope; keep an emergency stop within reach; check wiring and polarity before switching on; use a current-limited bench supply or a fuse; secure the robot so it cannot fall off the bench; wear safety glasses; tie back long hair and loose clothing; never work on live circuits. (4)

7.2 A user manual explains how to use the robot safely and correctly ✓✓. Any TWO contents ✓✓: safety warnings; setting up and switching on; operating instructions; specifications; maintenance and charging; fault-finding table; contact details for support. (4)

[8]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE